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Research on the transformation mechanism of carbon sequestration benefits of the digital economy in the wetlands around dongting lake under high-quality development: a case study of the dongting lake wetland reserve

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Abstract: At present, China's economy is at a critical stage of transition from high-speed growth to high-quality development, and the digital economy is playing an increasingly important role in driving the transition towards a green and low-carbon economy. Against the backdrop of the nation's pursuit of high-quality development, the digital economy is a key driver for achieving energy conservation and emissions reduction, carbon sequestration benefits, and improvements to the ecological environment. This paper takes the Dongting Lake Wetland Reserve as its study area and utilises panel data from the three cities of Yueyang, Changde and Yiyang covering the period 2005–2022. Using Stata software and a least squares regression model, the study focuses on examining the mechanisms through which the digital economy influences carbon emissions and the optimisation of the energy structure, as well as how it facilitates the realisation of carbon sequestration benefits. This study finds that the development of the digital economy in the Dongting Lake Wetland Reserve remains in its early stages and has not yet fully realised its positive role in regional energy conservation, emissions reduction and carbon sequestration benefits; however, an analysis using a one-period lag reveals that improvements in technological levels can enhance the positive effects of the digital economy on carbon emissions reduction and play a guiding role in promoting the transformation of the energy structure through high-quality economic development.

Keywords: High-quality economic development; the digital economy; carbon emissions; and energy structure reform.



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1. Introduction

The Chinese economy is currently at a critical juncture in its transition from high-speed growth to high-quality development. The introduction of the ‘dual carbon’ targets has established strict constraints for a comprehensive green transition in economic and social development, whilst the digital economy holds immense potential for improving resource allocation, enhancing energy efficiency, and promoting emissions reduction and carbon sequestration. Wetlands are among the world’s most important carbon sinks; their carbon sequestration rate per unit area is significantly higher than that of forests, and they play an irreplaceable role in maintaining regional climate stability. Dongting Lake is China’s second-largest freshwater lake and an internationally important wetland, situated within a vital ecological buffer zone in the middle reaches of the Yangtze River. The lake’s wetland vegetation covers a vast area, abundant peat reserves, and its carbon sequestration accounts for a significant proportion of the total carbon cycle within the entire river basin. However, due to prolonged and rapid development, the wetlands have suffered severe shrinkage, marked landscape fragmentation and ecological degradation. Against this backdrop, the key challenge for the region in jointly advancing ecological conservation and high-quality development is how to utilise the digital economy to transform the ecological value of wetland carbon sequestration into quantifiable and tradable economic benefits. This paper selects Yueyang, Changde and Yiyang—all located within the Dongting Lake Wetland Protection Area—as the subjects of study. Using panel data from 2005 to 2022, the paper employs the least squares method and mediation analysis to examine the impact pathways and moderating effects of the digital economy on carbon emissions (CE) and energy structure (ES). Given the difficulty in obtaining data at the prefecture-level city level, some indicators lack precision, and the model suffers from endogeneity bias; these issues will subsequently be addressed using a one-period lag method.

The impact of the digital economy on carbon emissions has recently become a hot topic of research. Existing literature suggests that the digital economy can promote the establishment of a clean, low-carbon energy system by optimising resource allocation and improving energy efficiency, amongst other means [1]. Some studies also indicate that the relationship between the digital economy and carbon emissions may follow an ‘inverted U-shaped’ pattern: that is, in the early stages of the digital economy’s development, driven by infrastructure construction, carbon emissions tend to rise; it is only once the application of technology reaches a certain level that its emission-reduction effects begin to gradually manifest [2]. Empirical analysis based on data from 269 prefecture-level cities indicates that the impact of the digital economy on green ecological efficiency exhibits non-linear characteristics [3]. The deep integration of digital technologies is transforming the ways in which services are delivered and value is created, whilst the realisation of the carbon neutrality vision hinges on systematic planning of technological pathways and breakthroughs in key areas [4]. With regard to international trade, the rapid development of the digital economy is reshaping the landscape of cross-border trade and business operations [5]. The impact of the digital economy on regional development disparities cannot be overlooked, as

there is significant heterogeneity between regions at different stages of development and with different characteristics [6]. In terms of innovation mechanisms, digital innovation is transforming the operational logic of organisations and the ways in which value is created, whilst system modelling techniques provide a methodological foundation for data analysis and decision-making governance in complex environments [8]. From the perspective of ecological security, the use of the DPSIR model for assessment effectively reflects the status and changes in regional ecological security [9]; from the perspective of industrial agglomeration, the impact of digital industrial agglomeration on the efficiency of the green economy has been preliminarily verified [10]; improvements in green technology R&D capabilities can amplify the emission-reduction effects of the digital economy, whilst insufficient R&D investment will affect the efficiency of technology transfer [1, 7]. Taken together, the existing literature has provided theoretical support and empirical evidence from various perspectives for understanding the relationship between the digital economy and carbon emission reductions; however, systematic examinations specifically targeting key ecological functional zones remain scarce, particularly with regard to regions such as the Dongting Lake area, which are characterised by both high ecological sensitivity and a late-developing economy. Most studies focus on examining the average effects of the digital economy, with few undertaking contextualised analyses that integrate the phased characteristics of the digital economy with actual regional development conditions; furthermore, there are very few studies that incorporate the transformation of wetland carbon sink functions into a unified analytical framework.

In contrast to existing research, the marginal contribution of this paper lies primarily in its focus on the wetlands protection area surrounding Lake Dongting. Based on panel data from 2005 to 2022, the study examines the impact of the digital economy and high-quality development on regional carbon emissions and energy structure in a phased manner. It reveals the phased characteristic of ‘initial carbon increase’ during the development of the digital economy in this region and enhances the reliability of the conclusions through a one-period lag robustness test. From a theoretical perspective, the impact of the digital economy on carbon emissions and energy structure is not static but varies according to the degree of digitalisation in different regions. In the early stages of development, both the industrialisation of the digital sector and the digitalisation of traditional industries must rely on substantial physical infrastructure, including the construction, investment and operation of 5G base stations, data centres, fibre-optic networks and associated power facilities. The entire process is inherently characterised by high energy consumption and high carbon emissions, with long payback periods and energy efficiency gains emerging only at a later stage. Furthermore, the industrial structure of the Dongting Lake region consists primarily of traditional manufacturing and resource processing industries. Small and medium-sized enterprises face practical constraints in their digital transformation due to high technical barriers and substantial modernisation costs; it will be difficult for them to generate emission-reduction dividends capable of offsetting the energy consumption of infrastructure in the short term. From the perspective of the energy mix, the increased demand for electricity at the outset of the digital economy is primarily met by the local thermal power system, which

remains dominant, which further slows down the process of increasing the share of non-fossil energy consumption. The moderating effect of high-quality development is phased; consequently, before the digital economy crosses a critical mass threshold, the guiding role of high-quality development in emission reduction pathways will be overshadowed by the energy consumption effects of infrastructure expansion. Both technological development and foreign trade will, in theory, be affected by the time lag between initial investment and subsequent output; emission reduction effects will only become apparent once these factors have accumulated to a certain extent.

2. Methods

This paper utilises panel data from the three cities of Yueyang, Changde and Yiyang, situated around Dongting Lake, covering the period 2005–2022 as the observational sample. Based on the full cycle from 2005 to 2022, the period from 2011 to 2022—characterised by the rapid development of the digital economy—is also selected, and group-comparison regression analyses are conducted separately for these two time periods; Furthermore, dependent variables, independent variables and control variables were selected, with the sources and calculation methods of each indicator clearly specified, to construct a regression model quantifying the direct impact of the digital economy and high-quality economic development on regional carbon emissions and energy structure transition; the least squares regression model was employed to understand the direction and significance of the core variables; simultaneously, to verify the reliability of the regression results, robustness tests were conducted using a one-period lag of the core independent variables. By cross-validating the baseline conclusions through grouped regression and one-period lag tests, the conclusions are rendered more persuasive. The results of the empirical analysis comprehensively present the descriptive statistics for each variable, the regression coefficients of the model, and the significance levels. The methodological framework is complete and can provide a referenceable empirical analysis paradigm for subsequent research on the same region and topic. The variables selected are as follows:

- Dependent variables: carbon emissions (CE), energy mix (ES)
- Independent variables: level of digital economic development (DE), level of high-quality economic development (HQ)
- Control variables: urban–rural consumption gap (GAP), industrial structure (LEV), technological development (TD), foreign trade (FT)

3. Results

3.1. Equations

$$\begin{aligned} CE_{it} &= \alpha_0 + \alpha_1 DE_{it} + \alpha_2 HQ_{it} + \alpha_3 X_{it} + \varepsilon_{it} \\ ES_{it} &= \beta_0 + \beta_1 DE_{it} + \beta_2 HQ_{it} + \beta_3 X_{it} + \varepsilon_{it} \end{aligned} \quad (1)$$

Here, i denotes a prefecture-level city and t denotes the year; the dependent variables are carbon emissions (CE_{it}) and energy structure (ES_{it}); the key explanatory variables are the

level of digital economic development (DEit) and the level of high-quality economic development (HQit); a series of control variables affecting high-quality economic development and related to digital economic development are denoted by X_{it} , including the urban–rural consumption gap (GAP) and industrial structure (LEV); the random disturbance term is ε_{it} , and α and β are the coefficients.

3.2. Tables

This paper conducts a descriptive statistical analysis based on data from the Dongting Lake region. As the 19th National Congress of the Communist Party of China in 2017 called for a ‘shift from rapid economic growth to high-quality economic development’, this paper divides the data for the Dongting Lake region into two time periods—2005–2022 and 2011–2022—for the purposes of analysis.

Table 1. Descriptive statistics for key variables, 2005–2022.

Variables	Mean	Standard deviation	Minimum value	Maximum value
CE	359.6	284.2	54.76	1,070
ES	0.156	0.0637	0.0447	0.305
DE	845,637	1.095e+06	8,234	4.027e+06
TFP	0.425	0.0679	0.276	0.575

a. The dependent variables are carbon emissions and energy mix; the key explanatory variables are the digital economy index and total factor productivity (which characterises high-quality economic development). b. Total factor productivity was calculated using the DEA-Malmquist index method. c. The sample size is 54 (3 cities $\times$ 18 years).

Table 2. Descriptive statistics for key variables, 2011–2022.

Variables	Mean	Standard deviation	Minimum value	Maximum value
CE	475.3	285.6	137.0	1,070
ES	0.155	0.0640	0.0447	0.305
DE	378,040	179,960	142,686	976,939
TFP	0.417	0.0648	0.276	0.575

a. The period from 2011 to 2022 represents a phase of rapid development in the digital economy, with a sample size of 36 (3 cities $\times$ 12 years). b. The average carbon emissions during this period (475.3) were approximately 32.2 per cent higher than the average for the

entire cycle (359.6), reflecting the pressure on energy consumption resulting from regional economic expansion.

a. The descriptive statistics in Tables 1 and 2 show that, over the period 2011–2022, the average values of indicators such as carbon emissions and the digital economy index in the Dongting Lake region exhibited an upward trend compared with the entire period from 2005 to 2022. The average carbon emissions rose from 3.596 million metric tonnes to 4.753 million metric tonnes, an increase of 32.2 per cent; This figure indicates that the region's economy is developing, the construction of digital infrastructure is accelerating, and energy consumption has consequently increased; the average total factor productivity fell slightly from 0.425 to 0.417, reflecting a relatively stable level of high-quality development over time. These changes provide a practical basis for further analysis of how the digital economy will impact carbon emissions and the energy structure, against the backdrop of improving national economic conditions and relevant policy adjustments.

Table 3. Results of the least squares regression.

Variable	Year 2005—2022		Year 2011—2022	
	CE	ES	CE	ES
DE	-0.000 (-0.74)	-0.000 (-0.30)	0.001*** (3.58)	- (-2.68)
TFP	-1023.012* (-1.85)	-0.018 (-0.12)	-544.005 (-0.79)	-0.192 (-1.08)
Constant term	1,512.092*** (4.38)	0.120 (1.32)	-128.681 (-0.23)	0.376** (2.63)
R-squared	0.306	0.036	0.417	0.224
Adjusted R ²	0.246	- 0.0478	0.334	0.113
Sample size	54	54	36	36

a. The figures in brackets represent the t-statistics. b. $p < 0.01$, $p < 0.05$, $p < 0.1$. c. The key explanatory variables are the digital economy index and total factor productivity.

Table 3 presents the results of ordinary least squares regression analyses conducted on data from the Dongting Lake Wetland Nature Reserve for the two time periods. Over the full period from 2005 to 2022, total factor productivity had a significant reducing effect on carbon emissions at a 90 per cent confidence level (coefficient of influence: 1,023.012, p-value less

than 0.05); however, its effect on the energy mix was not significant in either time period. This indicates that, viewed over the entire time scale, high-quality economic development has begun to demonstrate an initial positive role in driving carbon emission reductions, whilst the digital economy—still in its early stages of development—has yet to exert a statistically significant impact on carbon emissions or the energy mix. This is consistent with the fact that the digital economy has only experienced rapid growth in the last decade. During the period from 2011 to 2022, the coefficient of influence of the digital economy index on carbon emissions was 0.001, and it exhibited a positive and significant effect at the 1 per cent significance level; its impact on the energy mix was also significant. This implies that during the phase of accelerated growth of the digital economy, it notably drove up carbon emissions and hindered the transition towards a cleaner energy mix.

The results of the least squares regression indicate that, between 2011 and 2022, the digital economy index had a significant impact on both carbon emissions and the energy mix, whilst total factor productivity had no significant impact on either. Consequently, this paper conducts robustness tests solely on the impact of the digital economy on carbon emissions and the energy mix. This paper employs a one-period lag approach to test the stability of the digital economy's impact on carbon emissions and energy structure between 2011 and 2022, in order to mitigate potential endogeneity issues such as reverse causality and omitted variables.

Table 4. Results of the one-period lag robustness test.

Variables	(1)	(2)	(3)	(4)
	CE	L.CE	ES	L.ES
DE	0.001*** (0.000)		- 0.000** (0.000)	
L.DE		0.001*** (0.000)		-0.000*** (0.000)
TFP	-544.005 (685.897)	-1499.001* (811.336)	-0.192 (0.177)	-0.048 (0.193)
R-squared	0.417	0.351	0.224	0.288
Adjusted R ²	0.334	0.247	0.113	0.174
Sample size	36	33	36	33

a. Figures in brackets represent standard errors. b. $p < 0.01$, $p < 0.05$, $p < 0.1$. c. L. denotes the first-order lag term of the digital economy index. d. The sample size for the one-period

lag regression is 33 (3 cities $\times$ 11 years); the use of lag terms results in the automatic exclusion of data for 2011.

The results are shown in Table 4. both the Digital Economy Index and its first-order lagged term exhibit a positive relationship with carbon emissions at the 1 per cent significance level; their effects on the energy mix are also significant. Furthermore, the magnitudes and levels of significance of the coefficients for both variables are broadly consistent. This indicates that the impact of the digital economy on carbon emissions and the energy mix is stable over time; the results of the baseline regression have not been distorted by endogeneity issues; and the findings of the stability test are broadly consistent with and share the same nature as those of the baseline regression. this validates the conclusion that ‘the digital economy in the Dongting Lake Wetland Nature Reserve is still in the early stages of increasing carbon emissions’ as reliable, and further supports the assessment that the digital economy will drive carbon emissions in the short term.

4. Conclusion

This paper utilises panel data from the three cities of Yueyang, Changde and Yiyang for the period 2005–2022. It employs a least squares regression model to examine the direct impact of the digital economy and high-quality economic development on regional carbon emissions and energy structure, whilst conducting robustness tests using one-period lagged variables. The study indicates that, over the entire period from 2005 to 2022, the digital economy had no significant impact on either carbon emissions or the energy mix; during the phase of rapid digital economic development from 2011 to 2022, carbon emissions increased significantly and the digital economy had a dampening effect on the transition to cleaner energy. Robustness tests using one-period lagged variables also confirmed that this conclusion is stable and reliable; Overall, as the development of the digital economy in the Dongting Lake Wetland Protection Area remains at an early stage, there are phased contradictions between digitalisation efforts and the low-carbon transition. Consequently, the digital economy has not yet fully realised its positive role in regional energy conservation, emissions reduction and carbon sequestration. Furthermore, the impact of high-quality regional development in driving low-carbon initiatives has been relatively weak, failing to offset the increase in carbon emissions resulting from the development of the digital economy. Compounded by energy consumption pressures arising from factors such as the urban–rural income gap and the proportion of the secondary sector, this has further hindered the realisation of carbon sequestration benefits in the Dongting Lake Wetland; The results of the lag analysis indicate that improvements in technological levels can enhance the positive effects of the digital economy on carbon emission reductions and play a guiding role in promoting high-quality economic development and the transformation of the energy structure; at the same time, they demonstrate that the carbon reduction effects of digitalisation exhibit a significant lag. Currently, a majority of relevant literature confirms that the digital economy possesses long-term green emission-reduction value, complementing the findings of this study. It discusses the phased increase in carbon emissions characteristic of the etable heshiarly stages of

digitalisation in the Dongting Lake wetland ecological zone, thereby enriching research on regional heterogeneity between the digital economy and ecological carbon sequestration. However, this study has certain limitations: we have only discussed the direct effects of the digital economy and high-quality development, without introducing mediating variables or conducting mediating effect tests. Future research could expand upon mediating and moderating models to further refine the analysis of underlying mechanisms.

Declaration of generative AI and AI-assisted technologies

During the preparation of this manuscript, the authors used generative AI tools only to improve language and readability. The authors take full responsibility for the content of the manuscript.

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Authors' contribution

Conceptualization, Xiyu L; methodology, Xiyu L, Yuting Z, Yue Z and Weiqi D; software, Yuting Z; validation, Yuting Z, Yue Z and Weiqi D; formal analysis, Yuting Z; investigation, Yuting Z, Yue Z and Weiqi D; resources, Xiyu L; data curation, Yue Z; writing-original draft preparation, Yuting Z; writing-review and editing, Xiyu L; visualization, Weiqi D; supervision, Xiyu L; project administration, Xiyu L; funding acquisition, Xiyu L. All authors have read and agreed to the published version of the manuscript.

Conflicts of interests

The authors declare no conflict of interest.

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